Flight Dynamics of the Sikorsky UH-60A Black Hawk in JSBSim

An individual-blade rotor model after Sikorsky's GenHel, checked against US Army flight tests

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Abstract

A flight-dynamics model of the UH-60A Black Hawk for the FlightGear simulator has been written in JSBSim, replacing the parametric YASim helicopter model the aircraft used before. It follows the structure of the Sikorsky General Helicopter (GenHel) engineering simulation documented in NASA CR-166309: the four main rotor blades are simulated individually, each with flap and lead-lag freedom, a nonlinear lag damper and eight aerodynamic segments using the SC1095 tables over ±180° and Mach number, under a Peters–He finite-state dynamic wake; the tail rotor is built the same way, with GenHel's closed-form solution as the alternative, and the two are compared throughout; fuselage, stabilator and fin use GenHel's wind-tunnel tables, and a ring-vortex wake decides how much of the main rotor's wake reaches the tail in any wind, and puts the fin in the tail rotor's wake; the automatic flight control system (stabilator schedule, SAS, pitch bias actuator, flight path stabilization) and two T700 engines with the rotor speed degree of freedom complete it. The whole model, 1242 aerodynamic functions and 422 control-system components, is generated by a script from transcribed data, and runs in plain JSBSim without scripting. It was flown offline through a scripted programme and compared with US Army (USAAEFA) flight tests and with GenHel. Level-flight trims from 50 to 162 kt agree with the test within 9.0 % of longitudinal and 4.6 % of collective travel and 1.2° of pitch attitude. Out-of-ground-effect hover power is 0.949–0.979 of the tethered-hover test, rotor figure of merit is within 0.026 of the measured data, and ground-effect power savings are within 0.009. For the 17 control-step responses of NASA TM-88360 the model's root-mean-square rate error is below GenHel's in 11 of 16 usable cases (mean 2.10 against 2.77 deg/s). Hovering in winds from every direction, where GenHel's wash at the tail doubles the pedal's swing, the ring wake brings the pedal error from 0.77 to 0.41 in rms. The hover pedal step exposed a missing term: the rotor keeps its own angular momentum, so its speed relative to the airframe must follow the airframe's yaw; with that term the yaw-rate error after the step fell from 5.5 to 2.9 deg/s. Deficiencies are quantified: cruise power 2–10 % above one of two conflicting test sources, a weak pitch-up after an aft step at 100 kt (17.2 against 24.1 deg/s), an angle-of-attack derivative M_w a third of that of TM-85890's model, half the bank change between climb and descent that the aircraft shows, and a tail rotor that reaches its stall near the top speed, where the aircraft still had pedal to spare.

1 Introduction

The UH-60A distributed for FlightGear flies on YASim, whose helicopter model builds the rotor from a short list of geometric parameters and tuning factors. Its behaviour is set by tuning rather than by the aircraft's data, and its couplings, power characteristics and responses cannot be traced to measurements. For the UH-60A, unusually, the data exist in public: in the early 1980s Sikorsky documented its General Helicopter (GenHel) engineering simulation of the Black Hawk in full for NASA [1], NASA re-implemented and validated it against US Army flight tests [2, 3], and the Army's test reports [6, 5, 7] give trim, stability and performance data for the same airframes.

This report describes a JSBSim [21] model built on that material and how well it reproduces the tests. GenHel is used as the blueprint, not as gospel: where it leaves the physics incomplete (the induced power, Section 5.4) or where validation exposed an inconsistency, the model departs from it, and the departures are listed in Section 6. All of the flight model, including the rotor's integration, the control laws and the engines, is JSBSim XML; the FlightGear side only animates and instruments it. All result tables, and the numbers quoted from them, are written by the evaluation scripts.

2 Reference data

Table 1 lists the sources; most are scans, and every curve used was digitised from them. A few points of data reduction matter for the comparisons. *(i)* The control-step responses of TM-88360 were digitised from 68 plot pages (some scanned sideways or upside down); each case is replayed at its own test condition from table C1 of that report: weight, centre of gravity (CG), density altitude and speed. 8 of the cases, all five in hover among them, were flown at FS 359.2–359.6, near the aft limit; for the aft step at 100 kt this raises the pitch response by half. *(ii)* The trims of TM-88360 fig. B1 come from USAAEFA test 79-24; its own figure 1 shows that the three points plotted at 22–32 kt were flown at 3–15 KCAS, so they are near-hovers and are shown apart. *(iii)* Two tests of the same airframe, S/N 77-22716, disagree on cruise power: at 95 kt the main rotor torque of 79-24 (TM-88360) implies 4–6 % more power than the engine torque of 77-17, of which instrumentation drag explains about 2 %. *(iv)* The hover charts of the operator's manual represent the sixth-year aircraft with the external stores support system (ESSS) fairings, which needs about 3 % more hover power than the first-year aircraft; the model is compared with the latter.

3 The model

3.1 Implementation

A Python script writes the model from the transcribed tables (Table 2). JSBSim evaluates aerodynamic functions on demand, so the function graph must be free of cycles; every quantity that depends on its own past, the blade flap and lag angles and rates of both rotors, the wake states, the rotor speed, the tail rotor inflow, the blade's elastic twist and the engines' states, therefore lives in flight-control-system channels, which JSBSim runs once per frame in a fixed order and which can write properties back. These channels integrate the states explicitly at the simulation rate of 120 Hz, the rate FlightGear runs the model at: 12.9° of main rotor azimuth per step and 60° of the tail rotor's; the wake states use a semi-implicit step. Trims at this rate are within 0.8 % of those at 960 Hz and the

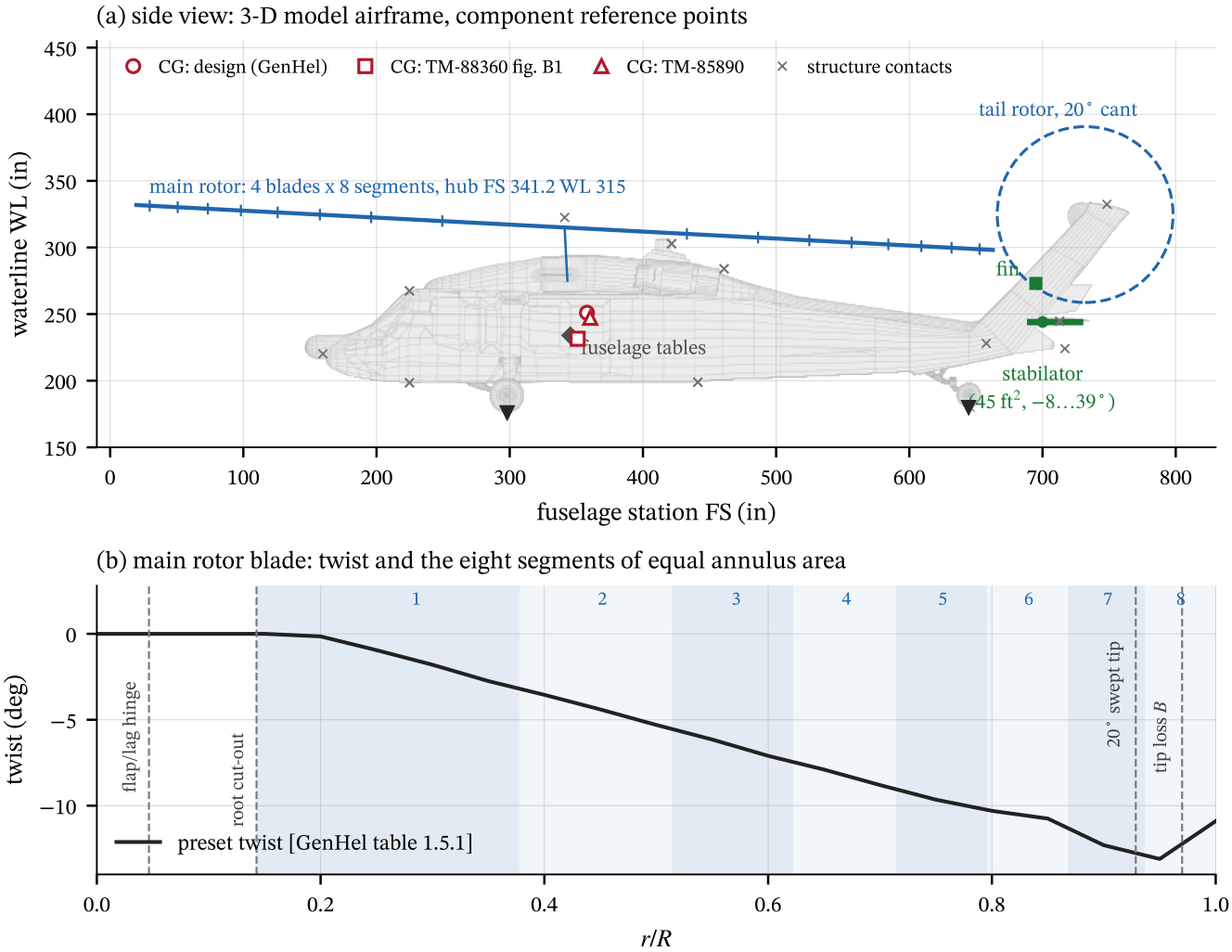


Figure 1: Geometry. (a) The FlightGear 3-D model's airframe in Sikorsky stations, with the main rotor disc (3° forward shaft tilt; ticks at the segment mid-points), the tail rotor, the stabilator and fin reference points, the reference point of the fuselage tables, the three centres of gravity used in this report, the wheels (▼) and the 17 structure contacts. (b) The main rotor blade: preset twist and the eight segments of equal annulus area.

peaks of control responses within 1–4 %; 480 Hz would be within 0.5 % at four times the cost of about 0.12 ms per step. The blade loads of a frame use the states of the previous frame. The same XML runs in FlightGear and in the offline harness used here.

3.2 Main rotor

Each blade is a rigid beam on a coincident flap and lag hinge at 4.66 % radius. Its eight segments of equal annulus area (Fig. 1b) see the air velocity at their mid-point from the full kinematics in shaft axes: the aircraft's translation and rotation, the rotor speed, the blade's flap and lag motion and the induced velocity. Lift and drag come from the SC1095 tables of GenHel against angle of attack (±180°; the drag of the reversed section near 180°, which GenHel's table sets to zero, is restored) and Mach number, which on the 20° swept tip is that of the velocity normal to the leading edge. Lift acts on the velocity normal to the span; drag acts along the whole flow including its spanwise part, which is what gives the blade's profile power its growth with advance ratio. A section drag increment of 0.0012 is added to the smooth wind-tunnel tables: a production blade's boundary layer is turbulent from near the leading edge, and tripping added about half of the NACA 0012 increment [10] to the SC1095 at full-scale Reynolds number [9]. GenHel's empirical torsional wind-up twists the tip nose-down in proportion to the mean blade load. The tip loss factor $B = 0.97$ removes the lift outboard of BR and, unlike GenHel, also enters the momentum balance (Section 5.4).

The flap equation of each blade contains the aerodynamic hinge moment, the hub's specific force, the gyroscopic terms from the body's rate and acceleration, the centrifugal stiffness (including the lead rate) and the lag damper; the blade rests on droop and anti-flap stops. The lag equation contains the aerodynamic and damper moments, the centrifugal stiffness of the hinge offset, the Coriolis moment of flapping and the rotor's acceleration. The lag damper's force comes from GenHel's force–stroke-speed table and acts through the geometry of its two pickups, one on the hub and one on the blade cuff, which turns with blade pitch, so that its flap and lag moments follow from virtual work. The blades' aerodynamic and inertial loads act on the hub at their hinges; the shaft torque drives the rotor speed.

Table 1: Sources and what they contribute.

Source	Content used
NASA CR-166309 [1]	GenHel: rotor, tail rotor, fuselage, empennage, controls, AFCS, engine equations; SC1095, fuselage and empennage tables; physical and mass data
NASA TM-85890 [2]	parameter tables, control rigging and mixing; stability derivatives of its own (simpler, rotor-map) model
NASA TM-88360 [3]	level-flight trims vs. airspeed (fig. B1); 17 control-step responses (app. D) with test conditions (table C1)
NASA CR-177360 [4]	GenHel against Army tests: sideslip sweeps; tail-rotor wash on the fin
USAAEFA 79-24 [5]	the test behind TM-88360: inherent sideslip, lateral-directional sweeps, static and long-term stability; fin and stabilator dimensions
USAAEFA 77-17 [6]	tethered hover, tail rotor shaft power, level-flight power (first-year aircraft); low speed in every wind direction (figs. 106–110)
USAAEFA 83-24 [7]	hover fits of 77-17 reanalysed, sixth-year aircraft with ESSS fairings
DTIC ADA180404 [8]	sideslip in climbs and descents
NASA TP-2003-212265 [9]	SC1095 geometry, tip sweep, drag at full-scale Reynolds number, tripping increment
LRTA paper [11]	UH-60 rotor figure of merit from flight and wind tunnel tests
JAHS 2004 paper [12]	accessory power; flat-plate drag area; tail rotor power of the Airloads flights
NASA TM 110362 [18]	hover derivatives identified from flight test; rotor speed equation
DTIC ADA181947 [19]	inherent sideslip with the ESSS fairings
USAAMRDL TR 72-67 [20]	dynamic stall delay (tail rotor experiment)
NASA TN D-234 [13]	ground effect in forward flight
METU thesis [14]	Peters–He finite-state wake equations
TM 1-1520-237-10 [15]	hover chart (ground effect, power), limits, procedures, AFCS functions, tail rotor construction
NASA T700 study [16]	engine and fuel control dynamics

3.3 Inflow

The induced velocity comes from the Peters–He generalised dynamic wake [14] with three axisymmetric states, giving the radial distribution in hover, and the first harmonic pair, driven by the rotor's hub moments and coupled to the wake skew. Its inflow gain matrices are tabulated against the wake skew parameter. The states are kept in the hub frame and turned into the frame of the in-plane flow for their update, so that a wake that swings round with the flow direction in a hover does not carry the states with it. Ground effect scales the induced velocity: in hover by $(1+0.3(R/Z)^2)^{-1/2}$, with Z the height of the hub, which reproduces the power savings of the operator's manual hover chart [15] (GenHel's own form gives half of them), and in forward flight by the ratio of forward-flight to hover interference of TN D-234 [13] against the angle of the wake. For use in the simulator, coloured noise can be added to the inflow and its gradients in the regimes where the real wake is unsteady (ground effect, translational lift, vortex ring); its intensities are estimates and it is off in all results here.

Table 2: The generated model: JSBSim system files, their content and the number of control-system components or named aerodynamic functions.

file	content	items
controls	rigging, mixing, servos, stabilator schedule, SAS, pitch bias actuator, FPS	122
rotor	blade flap and lag, lag dampers, Peters–He wake, tail rotor inflow, elastic twist, wake turbulence	145
drive	two T700 engines and their fuel controls, rotor speed, accessories, rotor brake	73
outputs	FlightGear outputs: rotor animation, instruments, vibration cue	82
aero	blade segments (4×8), tail rotor, fuselage, stabilator, fin, interference	1242

3.4 Tail rotor

Two tail rotor models can be built from the same data; both are compared in Section 5. Both are canted 20° so that the rotor also lifts, take GenHel's rigging from the pedals (with the collective mixing) to the pitch at the blade cuff, and see the same air at the hub.

GenHel's closed form. Bailey's solution for a rotor with linear twist [1] gives the thrust and torque coefficients in terms of the inflow and advance ratios, with pitch–flap coupling ($\tan\delta_3 = 0.70$) through the coning per pound of thrust, a tip loss of 0.92 and the inflow solved by a relaxed fixed-point step per frame over the effective disc. Its pitch is the cuff angle plus GenHel's 6° "blade pitch correction to linear twist": with the −18° twist, the linear blade has the cuff's pitch at a third of the radius. Lift stays linear in the angle of attack, without stall or reversed flow, and terms above μ^2 are dropped.

Blade-element rotor. The second model is built like the main rotor. Four blades flap on a hinge at the centre, for the bearingless crossbeam whose graphite spars run from tip to opposite tip [15]; their flap inertia follows from the Lock number (3.38) and a weak spring stands for the spars' stiffness, which is not published (flap frequency 1.02 per revolution, in keeping with GenHel's coning per pound of thrust). Each blade's pitch is the closed form's, less $\tan\delta_3$ times its own flapping. Six segments from $0.2\,R$ use the SC1095 tables against angle of attack (±180°) and Mach number in the air at the hub plus the blade's own motion and a uniform dynamic inflow (Pitt–Peters momentum over the effective disc), which reverses with the thrust and is not limited in the vortex ring state. The thrust, in-plane forces and hub moments, the drive torque among them, come from the blades.

Air at the tail rotor. GenHel computes it "in the same manner as the empennage" [1]: with the stabilator's and fin's main rotor wash factors, which depend on the wake's skew and the disc's tilt but not on the direction the wake is blown, so that the far wake's twice-mean induced velocity reaches the tail rotor in any wind. The tail rotor hub is $1.21\,R$ behind the main rotor's and 0.8 ft above it: at the upper edge of the wake when that is blown aft, clear of it when it is blown sideways or forward. The model takes the main rotor's induced velocity at the tail rotor from a skewed cylinder of ring vortices (uniform loading, the wake of TN D-234 [13]) leaving the

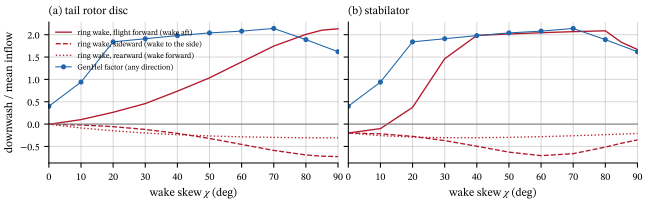


Figure 2: Main rotor downwash, over its mean inflow, (a) over the tail rotor's disc and (b) over the stabilator: the ring wake for the aircraft flying forward, sideward and rearward (level disc, $a_1 = 0$), and GenHel's interference factor, which applies in any direction.

tip-path plane, averaged over the tail rotor's disc and tabulated against the skew, the direction of the in-plane flow and the longitudinal flapping (Fig. 2a). The fuselage's sidewash and the dynamic-pressure loss of the fin maps are GenHel's.

Fin blockage. The tail rotor is a tractor whose wake meets the fin. GenHel multiplies the thrust by 0.796 in the hover, fading to 1 by 30 kt in any wind direction. Here the tail rotor's own ring wake, skewed by the flow through and along its disc, gives the induced velocity over four spanwise strips of the fin (the planform of the principal dimensions [5]), and the blockage is the fin's own lift and drag in it: largest in the hover and with the wind from the right, which blows the wake onto the fin; small with the wind from the left, which blows it back through the rotor; a few per cent of the thrust in cruise. The ring wake puts 13.6 % of the hover thrust on the fin, where Sikorsky measured 20.4 % (GenHel's 0.796), and it is used as it is. Scaled up to Sikorsky's figure (by 1.25 in velocity) the jet at the fin, only 0.2 R behind the disc, would be close to its far-wake speed, and the fin would give back more of the rotor's yaw damping than the rotor loses to it (Section 5.7).

3.5 Airframe

The fuselage forces and moments come from GenHel's tail-off wind-tunnel tables against angle of attack and sideslip, evaluated with the local flow that includes the main rotor wake through GenHel's interference factors (functions of the wake skew and the disc's tilt). In rearward flight the tables are mirrored front to back. The stabilator and fin are strips with their own lift and drag tables over ±180°, the rotor wake, the fuselage's downwash, sidewash and dynamic-pressure loss at the tail. At the stabilator the ring wake reproduces GenHel's downwash factor, measured with the wake blown aft, from 40° of skew up (Fig. 2b); GenHel's factor is kept and the ring wake adds how the downwash changes when the wake is blown sideways or forward. The fin's in-plane wash comes from the ring wake over its planform. GenHel's empirical downwash correction terms [1], a side force, a rolling moment and a sideslip-dependent pitching moment for the wake's swirl at the empennage, from rotor-on wind tunnel tests and flight test, complete the airframe.

3.6 Flight controls

The rigging and the mechanical mixing (collective to cyclic and pedal, pedal to longitudinal cyclic) follow TM-85890 [2]; the primary servos are second-order systems. The automatic flight control system is GenHel's: the stabilator scheduled on airspeed and collective with pitch rate and lateral acceleration feedback; the two stability augmentation channels (SAS 1 analogue, SAS 2 digital, 10 % authority together); the pitch bias actuator; and the flight path stabilization (FPS) with attitude, airspeed and heading hold and turn coordination through the trim actuators. FlightGear cannot move the pilot's stick, so the FPS moves the stick's trim point instead, and the pilot's trim release becomes automatic when a control moves away from it. The tail rotor rigging carries 0.5 in of left pedal that GenHel's rigging lacks: TM-88360 found GenHel "a nearly constant pedal bias of approximately 0.5 in. to the right" of the flight test in level flight. CR-177360 obtained a similar shift at 140 kt by putting 1.2 times the tail rotor's induced velocity on the fin, but judged the factor unsubstantiated; with the fin in the tail rotor's ring wake the model finds that interaction at a few per cent of the thrust in cruise, so the offset is kept as rigging.

3.7 Engines, drive and mass

Each T700-GE-700 is modelled after GenHel's engine module and the NASA integration study [16]: the hydromechanical unit meters fuel within acceleration and deceleration schedules to hold a gas generator speed reference from the power lever or the collective's load-demand spindle, trimmed by the power-turbine speed governor; the gas generator follows with its speed-dependent lag and the power turbine's torque depends on both speeds. Freewheel clutches connect the engines to the main gearbox; the rotor speed is a degree of freedom with the inertia of hub, drive and blades. The blades' share, 6970 slug ft^2 about the shaft (a fifth of the airframe's yaw inertia), keeps its own angular momentum: the rotor's speed relative to the airframe changes by the airframe's yaw acceleration about the shaft [18], so a yaw droops or speeds the rotor, the tail rotor with it, and the governor's answer reaches the airframe as the drive torque's reaction. The gearbox absorbs 3 % of the main rotor torque (assumed) and the accessories 65.8 hp, Sikorsky's figure [12]. Table 3 lists the principal data. The default loading is GenHel's design point; each validation case uses its own test loading. The landing gear has two main wheels and a castering tail wheel; 17 structure contacts taken from the 3-D model's skin carry the aircraft when it lands on its belly or rolls over.

4 Method

The evaluation drives JSBSim from Python with the same files as FlightGear, at 120 Hz, with the wake noise off. A trim solves for the four controls and the attitudes that zero the six accelerations averaged over two rotor revolutions (Hann-weighted, so that the tail rotor's 4.6 revolutions per main rotor revolution leave no phase-dependent remainder), with the airframe held while the rotor and wake settle. Away from the nose the trim variables are the pitch and roll attitudes with the velocity fixed in the earth frame, since angle of attack and sideslip lose their meaning at 90° off the nose. Atmospheres reproduce each test's pressure and density altitude (JSBSim's temperature offset and the geometric altitude are iterated). The programme:

- *Level-flight trims* from hover to 162 kt at the loading of TM-88360 fig. B1 (16 000 lb, FS 351, WL 231.5, 5250 ft);
- *lateral-directional trims*: zero-sideslip and wings-level trims at 60, 100 and 140 kt, in 1500 fpm climbs and de-

Table 3: Principal data.

Main rotor [1]	
radius; blades; chord	26.83 ft; 4; 1.73 ft
solidity; Lock number (sea level)	0.0821; 8.1
speed (100 %); tip speed	27.0 rad/s; 724 ft/s
flap/lag hinge offset; aerodynamic root	1.25 ft (4.7 %); 3.83 ft
blade weight; flap inertia	256.9 lb; 1513 slug ft^2
airfoil; twist; tip	SC1095 (GenHel tables); −18° nonlinear; 20° sweep outboard of 0.929 R
segments	8 of equal annulus area, mid-points 0.29…0.97 R
tip loss B; section drag increment	0.97; 0.0012
shaft tilt; swashplate phase	3° forward; −9.7°
lag damper	nonlinear force–speed table [1], pickup kinematics
Tail rotor [1, 2]	
radius; blades; solidity	5.5 ft; 4; 0.1875
speed; cant; tan δ_3	124.62 rad/s; 20°; 0.70
twist; tip loss; fin blockage	−18°; 0.92; 0.796 below 30 kt
Airframe [1, 3]	
design gross weight; CG	16001 lb; FS 358, WL 251
inertias $I_{xx}, I_{yy}, I_{zz}, I_{xz}$	5462, 39469, 36887, 1574 slug ft^2
stabilator; fin	45 ft^2, −8 . . . +39° TED; 32.3 ft^2
Drive [1, 12]	
engines; transmission	2 × T700-GE-700, 1553 shp; 2828 shp
rotor inertia (hub, drive, blades)	7911 slug ft^2
accessories; gearbox loss	65.8 hp; 3 % of main rotor torque

scents and with the CG 2.1 in right and 4.1 in left, against the digitised sweeps of 79-24 and ADA180404;

- *power*: out-of-ground-effect hover from 11 000 to 22 000 lb at the 2302 ft test site; level flight at $C_T \times 10^4 = 65$ to 90 and $\mu = 0.10$ to 0.32; the isolated rotor's figure of merit at sea level; and blade-element momentum calculations of the same blade as a reference;
- *ground effect*: the test autopilot (written for the FlightGear flight tests) hovers the model at 5, 10, 20, 40 and 100 ft wheel height;
- *control responses*: the recorded control increments of the 17 cases of TM-88360 app. D, applied open loop from a trim at the test condition with the SAS, FPS and pitch bias actuator off and the stabilator fixed, as in the test;
- *stability derivatives* at 100 kt at the loading of TM-85890 (16 400 lb, FS 360.4), by central differences of the aerodynamic forces with the airframe held;
- *low speed in every wind direction*: trims at 10, 20, 30, 35 and 40 KTAS and relative wind azimuths from 0 to 360° at 25 ft wheel height, at the conditions of the two aircraft of 77-17 figs. 106–110 (16 880 lb at 9860 ft and 16 330 lb at 2080 ft density altitude, forward CG; weights from the plotted C_T), against their pedal, longitudinal cyclic and tail rotor shaft power;
- *tail rotor power in level flight* against the UH-60A Airloads flights 84–90 [12] at main rotor weight coefficients C_W = 0.0065 to 0.0091, wings level as flown and at 5° of nose-left sideslip, 16 500 lb at the density giving each C_W;
- *hover yaw derivatives*, quasi-static and by component, at the hover of the pedal step and at the 1-kt condition of TM-85890.

Each tail-rotor-related stage is run with both tail rotor models, and the crosswind trims also with GenHel's wash factors at the tail.

Table 4: Level-flight trim: model minus test (TM-88360 fig. B1); power as model/test −1. *Flown at 3–15 KCAS.

kt	lon	lat	coll	ped	θ	ϕ	power
		% of travel			deg		%
22*	−0.4	−5.1	−7.3	+1.2	+1.3	−1.1	−12
30*	−2.0	−13.2	−11.7	+3.0	+0.3	+1.5	−24
32*	−2.6	−12.9	−14.5	+5.7	+0.5	−0.4	−27
50	+3.5	−4.5	+3.1	−1.7	+0.3	−0.3	+9
72	+2.6	−1.7	+2.7	−4.3	−0.1	−0.2	+2
95	+1.7	+0.1	+1.3	−3.0	+0.5	−0.6	+2
117	+3.7	+0.9	−0.6	−2.8	−0.6	−1.0	−2
139	−0.3	+3.5	−4.6	−1.1	+0.5	−0.6	−4
162	−9.0	+9.5	−4.3	−13.2	+1.2	+1.4	+2

5 Results

5.1 Level-flight trim

Figure 3 and Table 4 compare the trims. From 50 to 162 kt the longitudinal cyclic is within 9.0 % of its travel, the collective within 4.6 %, the pitch attitude within 1.2° and the roll attitude within 1.4°. Main rotor power is within −4 to +9 %, the largest excess at 50 kt; the model reproduces the test's 930–950 hp minimum between 50 and 95 kt only from 72 kt up. The pedal is −4.3 to −1.1 % of travel from the test between 50 and 139 kt with the blade-element tail rotor and −6.7 to −1.9 % with Bailey's. The largest differences are at 162 kt: the collective 4.3 % low, the lateral cyclic 9.5 % right of the test, the nose 1.2° high, and the pedal −13.2 % from the test, which moves 7.5 % to the right between 139 and 162 kt. That move is the collective-to-pedal mixing: the collective rises by a fifth of its travel, and through the rigging the test's tail rotor pitch goes from 15.2 to 16.3° (+1.1°), against 14.8 to 17.9° (+3.1°) for Bailey's rotor (−9.3 % of pedal from the test). The blade-element rotor, asked for some 1600 lb at an advance ratio of 0.4, works past the static stall of its retreating blades, which begins near 17.5°: at 120 Hz it reaches the balance at 19.5° (+4.8° from 139 kt), with the cyclic and the pedal the furthest from the test of any speed; at 480 Hz, with 15° of its azimuth per step instead of 60°, more pitch no longer adds thrust and the trims stop short of a balance from about 158 kt. The point is at the edge of what the rotor's step resolves. Its sections pitch there at a reduced frequency near 0.2, where real sections stall later. Gormont's dynamic stall delay [20], built in as a switch, leaves the trims below the stall unchanged. With its reattachment delay applied once a section has stalled, the limit stays where it is; without it the 162 kt point trims, at 19.3° and 1780 lb, the main rotor at 2600 hp against the test's

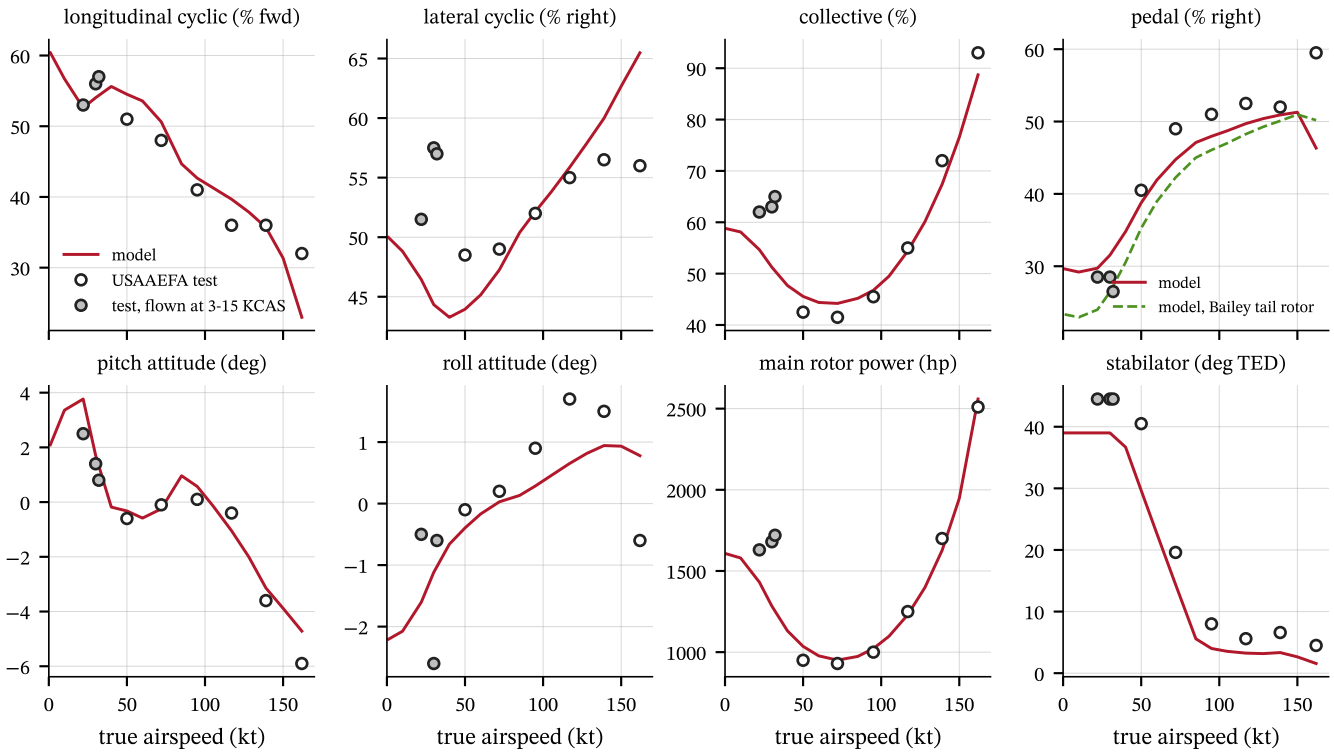


Figure 3: Level-flight trim against the USAAEFA test of TM-88360 fig. B1: 16 000 lb, FS 351, 5250 ft density altitude. Grey symbols: points plotted at 22–32 kt but flown at 3–15 KCAS. Controls are in per cent of travel; power is the main rotor's.

2510. The crossbeam's flap stiffness, which is not published, also moves the limit (with a flap frequency of 1.15 per revolution instead of 1.02 the point trims). Over the scatter of the pedal readings (about 1° of tail rotor pitch) the model follows the test's pitch from 95 to 139 kt; the test's small rise to 162 kt, with the main rotor's torque up by half, points to anti-torque the model lacks at the top speed. The near-hover test points need 7–14 % more collective than the model at 22–32 kt and 3–4 % more than its hover at 1 kt; there the stabilator's schedule saturates at the model's 39° limit, where the test reads 44.5°.

5.2 Lateral-directional trim

At zero sideslip (Table 5) the model's lateral cyclic is within 0.32 in of the test and its bank within 1.4°. GenHel's downwash correction terms matter here: without them the model trimmed at up to 2.5° of left bank, a side force of some 900 lb missing near the CG at 100 kt, which neither the tail rotor nor the rotor's tilt can supply. Two effects are only partly reproduced: the bank swings by 2.3° between a 1500 fpm descent and a climb against 4.3° in the test, and moving the CG from BL −4.1 to +2.1 in costs −0.98 in of lateral cyclic against −1.48 in, so the model's roll moment per inch of cyclic is about 1.5 times too large, or the test's lateral CG shift smaller than its loading implies.

5.3 Low speed in every wind direction

Hovering in a wind is where the tail rotor works hardest and where its air is least like the free stream. In the 77-17 tests (Fig. 4, Table 6) the aircraft needs the most left pedal with the wind from the right front, its critical azimuth, where only 0.5 in of pedal remained at 35 KTAS, and the most right pedal with the wind from the left; its tail rotor takes 340 hp at 60° and 25 hp at 270°. With GenHel's wash factors at the tail the model has this shape but swings nearly twice as far, and runs out of left pedal at 35–40 KTAS: its tail rotor meets the far wake's downwash and, in sideward flight, a crosswind 1.4 times the real one. With the ring wake at the tail and the fin in the tail rotor's own wake, the pedal error falls from 0.77 to 0.41 in rms with the blade-element rotor (0.36 in with Bailey's) and the longitudinal cyclic error from 0.68 to 0.49 in. The blade-element rotor keeps pedal in reserve throughout, 0.4 in at the least, as the aircraft did; Bailey's, less effective per degree, runs out at 40 KTAS with the wind from the right front. With the fin's blockage as the ring wake gives it (13.6 % of the hover thrust, Section 3) the blade-element rotor's pedal lies 0.42 in right of the 716 aircraft's on average, the high-altitude one. The tail rotor's shaft power is 59 hp rms from the test's, lowest against it with the wind from the right front; the pedal remains furthest to the right of the test with the wind from behind and from the left rear.

5.4 Power

GenHel applies the tip loss factor to the blade lift only. Its inflow is the momentum solution for the full disc, so the induced power it computes is ideal. The physical counterpart of the lost lift near the tip is a wake carried by the effective disc B^2A [17]: the induced velocity is $1/B$ times the full-disc value in hover and $1/B^2$ edgewise. The model applies this to both rotors by scaling the Peters–He forcing, and the tail rotor's momentum term, by $1/B^2$. The isolated rotor's figure of merit (Fig. 5d) then lies within 0.010 of a blade-element calculation with annulus momentum and Prandtl's distributed tip loss, and within 0.026 of the measured values; with the lift cut alone it is 0.028 higher.

The complete aircraft (Fig. 5a, Table 7) needs 0.949–0.979 of the measured engine power in the range of the test data and

Table 5: Lateral-directional trim against USAAEFA 79-24 sweeps (zero sideslip, from the sweep through it) and ADA180404. Lateral cyclic in inches from full left, pedal in per cent of travel from full left.

	roll (deg)		lat. cyclic (in)		pedal (%)	
zero sideslip	test	model	test	model	test	model
60 kt level	0.1	−0.2	4.84	4.51	43	42
100 kt level	1.8	0.4	5.48	5.29	49	48
140 kt level	1.4	1.0	5.87	6.02	51	51
100 kt, 1500 fpm descent	3.2	1.5	–	4.96	49	50
100 kt, 1500 fpm climb	−1.1	−0.9	–	5.48	49	44
100 kt, CG BL +2.1 in	2.5	1.0	4.60	4.96	–	48
100 kt, CG BL −4.1 in	−0.7	−0.8	6.08	5.94	–	50

Wings level, sideslip (deg, + wind from the right): model 60 kt +1.0, 100 kt −1.0, 140 kt −1.3; 100 kt climb +2.4, descent −4.1. Test: −4 at 67 KCAS to −1 at 150 KCAS [5]; +5 in IRP climbs, −5 in 1000 fpm descents [8].

Table 6: Low speed in every wind direction (77-17 figs. 106–110, 10–40 KTAS; aircraft 716 at 9860 ft, 717 at 2080 ft): model minus test over all azimuths and speeds, with GenHel's wash factors at the tail, and with the ring wake and either tail rotor.

	GenHel wash		blade-element		Bailey	
	mean	rms	mean	rms	mean	rms
pedal 716 (in)	+0.56	0.84	+0.42	0.48	+0.01	0.27
pedal 717 (in)	+0.14	0.69	+0.10	0.34	−0.22	0.43
long. cyclic 716 (in)	+0.16	0.57	+0.47	0.59	+0.22	0.42
long. cyclic 717 (in)	−0.61	0.76	−0.28	0.39	−0.48	0.56
TR power 716 (hp)	−43	57	−38	59	−42	64

Table 7: OGE hover: model against the first-year aircraft's tethered-hover fit [7]; tail rotor against 77-17 fig. 10 (bracketed: outside its data).

weight	C_T	$C_P \times 10^5$		ratio	tail rotor
lb	$\times 10^4$	model	test		ratio
11 000	43.5	36.80	36.57	1.006	(0.77)
13 000	51.4	43.74	44.69	0.979	(0.76)
15 000	59.3	51.47	53.46	0.963	(0.76)
17 000	67.3	60.00	62.83	0.955	0.77
19 000	75.2	69.27	73.00	0.949	0.79
21 000	83.1	79.53	84.76	0.938	0.83
22 000	87.0	85.30	91.30	0.934	0.86

0.934 at 22 000 lb; the deficit grows with thrust. The tail rotor needs 0.77–0.86 of its measured shaft power. In level flight (Fig. 5c) the model needs 3 to 11 % more power than the 77-17 data, 10 % at $\mu = 0.22$, while its main rotor power agrees with the 79-24 measurements of the same airframe (Table 4) within 6.4 % from 72 kt up once the latter's instrumentation drag (2.8 ft^2 [12]) is allowed for. The power budget (Table 8) shows nothing out of the ordinary: the airframe's drag area of 25.8 ft^2 matches Sikorsky's 25.7 ft^2 [12], induced power factors are 1.06 in hover and 1.12 in cruise, and the blades' profile drag is 43 % of main rotor power at $\mu = 0.22$, from section drag coefficients that agree with the SC1095 polars [9]. The pattern that remains, too little power in hover and too much in cruise against 77-17, survives any choice of the drag increment; it is the largest unexplained item of the model. The hover download is 2.4 % of weight, for which no UH-60 measurement was found.

Table 8: Power budget: OGE hover at 17 000 lb (2302 ft site) and level flight at $C_T = 0.0070$ (5250 ft).

	hover	μ 0.22	μ 0.30
weight (lb)	17000	16886	16886
main rotor, induced (hp)	1356	406	301
blade profile drag	347	458	658
propulsive (airframe drag)	0	205	502
tail rotor	177	60	72
gearbox and accessories	117	98	110
engine shaft power (hp)	1998	1226	1642
induced power factor κ	1.06	1.12	1.10
airframe drag area (ft^2)	–	25.8	25.4
download (% of weight)	2.4	–	–

5.5 Tail rotor power in level flight

The Airloads flights measured the tail rotor's power from the intermediate shaft torque (Fig. 6). Here the model needs more than the test: from $\mu = 0.12$ to 0.20, $0.79{\times}10^{-3}$ against $0.61{\times}10^{-3}$ (Bailey's rotor 0.83), and in cruise, $\mu = 0.22$ to 0.32, $0.64{\times}10^{-3}$ against $0.31{\times}10^{-3}$, where the measured value lies below what the blades' profile drag alone should take ($\approx 0.3 \times 10^{-3}$ for a mean section drag coefficient of 0.0095). From $\mu = 0.33$ up the measured power rises steeply; Bailey's rotor follows it at 1.1 times, while the blade-element rotor reaches its thrust limit at the heavier weights, as at 162 kt. The excess in forward flight is the opposite of the hover, where the model's tail rotor needs less than the 77-17 aircraft; the Airloads aircraft was a sixth-year one with the ESSS fairings. The comprehensive analysis of the same flights [12], with a free wake, also exceeds the measurement in cruise and comes closer with 5° of nose-left sideslip. The flights were flown wings level, which takes some sideslip, and the ESSS fairings add 2–3° of nose-left inherent sideslip to the basic aircraft's [19]; the model, wings level, has about 1°. At 5° the tail rotor moves into its own wake at some 18 ft/s and the fin takes part of its load: the model's cruise value falls from 0.64 to $0.50{\times}10^{-3}$, nearly half the way to the test's $0.31{\times}10^{-3}$. The rest remains open. At $C_W = 0.010$ and 0.011 the model does not trim beyond $\mu \approx 0.2$, the sections at their static stall; those points are left out.

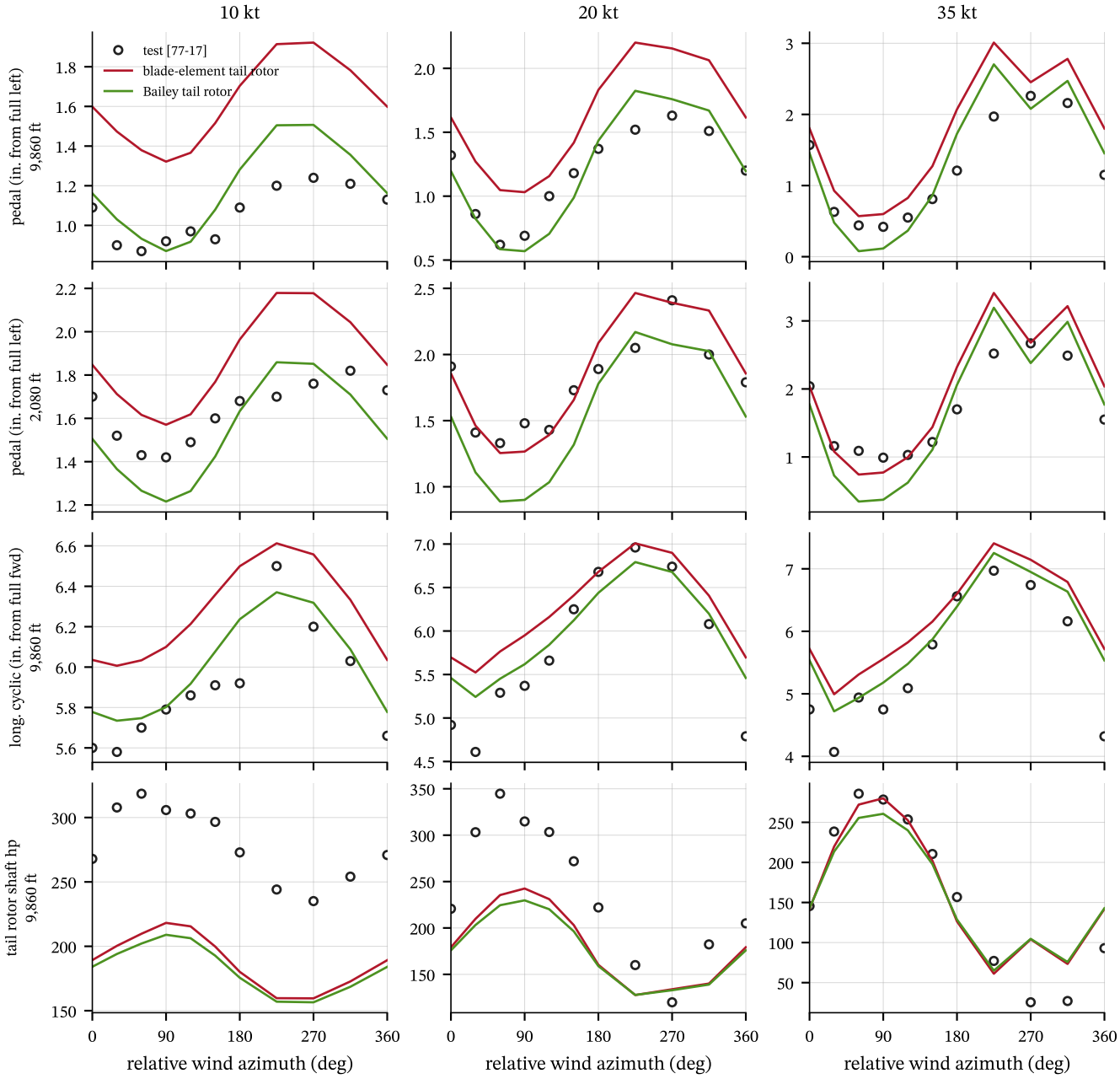


Figure 4: Low speed in every wind direction, 25 ft wheel height, against USAAEFA 77-17 figs. 106–110: pedal of both aircraft (716 at 9860 ft, 717 at 2080 ft density altitude), longitudinal cyclic and tail rotor shaft power of 716, with both tail rotor models. Relative wind azimuth from the nose, clockwise: 90° is the wind from the right (right sideward flight), 180° rearward flight.

Table 9: Ground effect: hover torque relative to 100 ft wheel height.

wheel height (ft)	model	flight manual	difference
5.0	0.809	0.800	+0.009
9.9	0.869	0.860	+0.009
19.9	0.932	0.925	+0.007
39.9	0.976	0.972	+0.004

5.6 Ground effect

With the test autopilot holding the height (Fig. 7, Table 9), the torque ratios are within 0.009 of the operator's manual at every height; flights in FlightGear with the same autopilot recorded 0.81, 0.87, 0.93 and 0.98.

5.7 Control response

Figure 8 shows six of the 17 cases and Table 10 all of them. The model's rms error is below GenHel's in 11 of 16 cases, with a mean of 2.10 deg/s against GenHel's 2.77 deg/s. In hover the model sits between GenHel and the test: GenHel's peak roll and pitch rates after cyclic steps are 38–106 % larger than the test's, the model's 2–41 %. The model's peak yaw rate after the pedal step in hover is 16 % larger than the test's (rms error 2.9 deg/s, GenHel 2.5). That case found a missing term: the rotor speed was integrated relative to the airframe from the torques alone, so a yaw left the rotor's speed relative to the airframe unchanged and the rotor's inertia, a fifth of the airframe's, out of the yaw axis. With the airframe's yaw acceleration in the rotor speed's equation (Section 3) a yaw to the left droops the rotor and the tail rotor with it, and the governor's answer reaches the airframe as the drive torque's reaction; the hover pedal step's error fell from 5.5 to 2.9 deg/s.

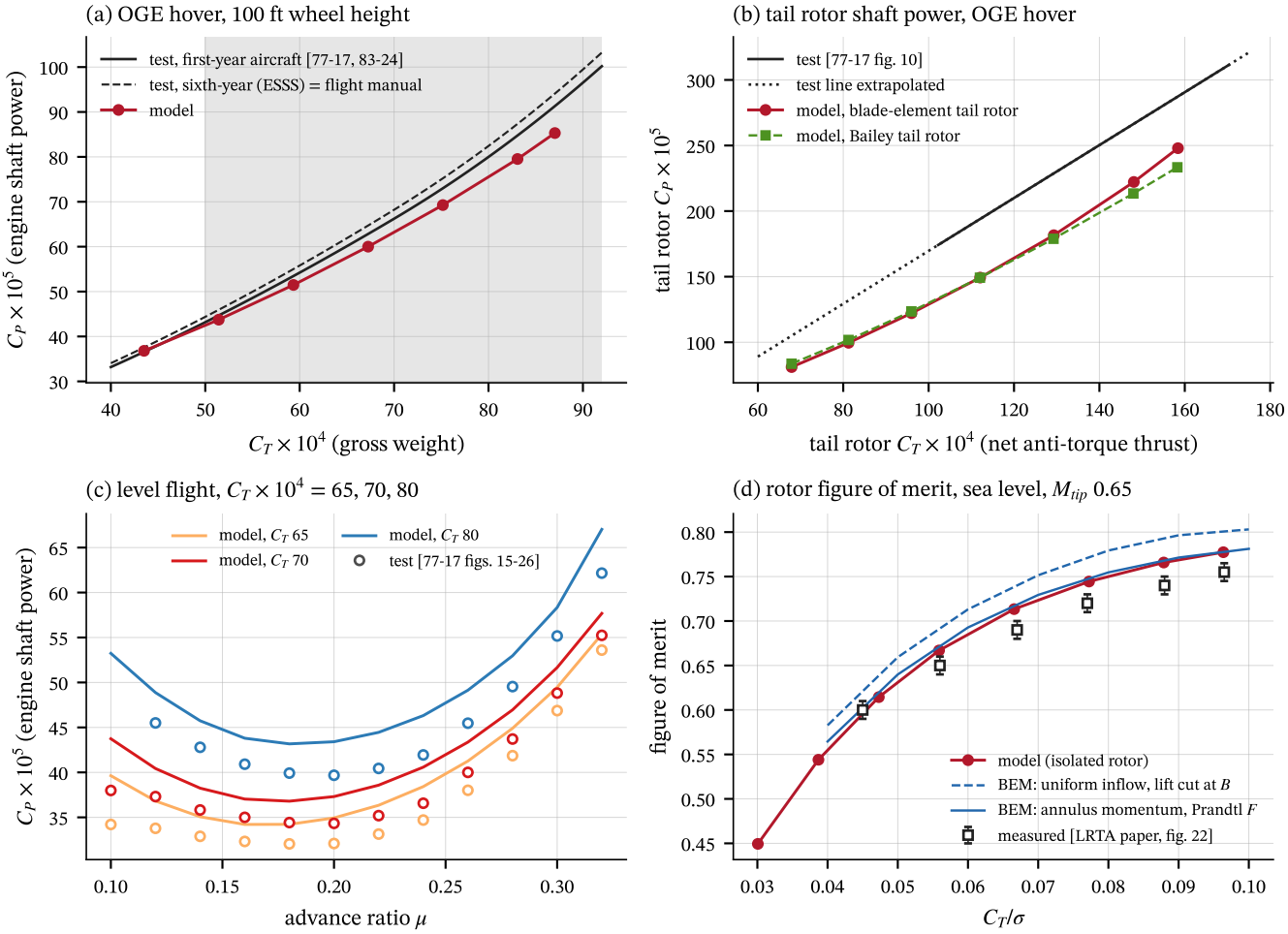


Figure 5: Power. (a) Out-of-ground-effect hover at the 2302 ft test site against the tethered-hover fits of 83-24 (shaded: test data range). (b) Tail rotor shaft power in the same hovers against 77-17 fig. 10. (c) Level flight against the faired data of 77-17 figs. 15–26 at 263 referred rpm. (d) Figure of merit of the isolated rotor, with two blade-element momentum (BEM) calculations of the same blade.

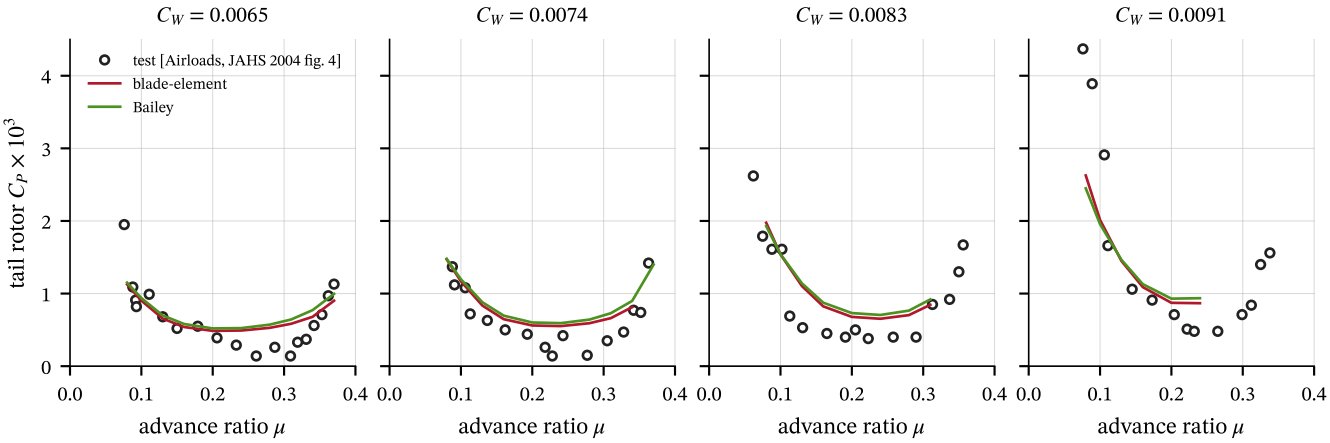


Figure 6: Tail rotor shaft power in level flight, wings level, against the UH-60A Airloads flights at four main rotor weight coefficients, with both tail rotor models. C_P of the tail rotor on its own disc and tip speed.

Held in the hover, with rotors and governor settled, the model's yaw damping is $N_r = -0.24\,\mathrm{s}^{-1}$: −0.18 from the tail rotor, −0.07 from the main rotor's torque, which rises with the rotor's speed through the air, and +0.02 from the fin, whose share of the tail rotor's jet grows as the tail swings into it. At the 1-kt condition of TM-85890 it gives $N_r = -0.25\,\mathrm{s}^{-1}$, N_v = 0.0076 and 0.53 rad/s^2 per inch of pedal, against −0.334, 0.0098 and 0.604 for that report's model. Flight identification of the UH-60A in hover gave $N_r = -0.445\,\mathrm{s}^{-1}$ [18], and a first-order fit of the 209 test $-0.49\,\mathrm{s}^{-1}$ for 27 deg/s^2 per inch (model −0.31 and 26, GenHel −0.32 and 21). The test's yaw acceleration, though, falls in the first half second and then stays nearly constant for two seconds, and after the collective step of 212 the yaw rate keeps rising: shapes that no first-order damping makes. The sideward-flight trims of 77-17, which test the same tail rotor physics in the steady state (Section 5.3), agree with the model. The hover yaw damping is left open between these readings. The clearest deficiency is the aft cyclic step at 100 kt: the test reaches 24.1 deg/s of pitch rate, GenHel 19.3 deg/s, the model 17.2 deg/s. At 100 kt both

Table 10: Control responses (TM-88360 app. D): peak on-axis rate in the 2.5 s after the input, and the rms error of the rate against the test over that interval (bold: model below GenHel). †The test pitch-rate trace of 509 could not be digitised reliably; excluded from the statistics.

				peak (deg/s)			rms error	
file	condition	input		test	GenHel	model	GenHel	model
201	hover	1 in right cyclic	p	9.8	13.5	10.0	2.4	**2.2**
202	hover	1 in left cyclic	p	−13.1	−20.7	−15.3	6.5	**1.4**
204	hover	0.5 in forward cyclic	q	−5.5	−11.4	−7.8	3.5	**2.0**
212	hover	1 in up collective	r	13.8	9.0	5.8	2.0	3.8
209	hover	1 in left pedal	r	−35.0	−31.9	−40.4	2.5	2.9
504	60 kt	1 in left cyclic	p	−10.5	−12.1	−9.8	1.3	1.5
509	60 kt	1 in aft cyclic	q	−26.8†	14.5	−14.7	13.2	15.8
405	60 kt	0.5 in down collective	r	0.9	1.0	−1.0	0.5	1.3
510	60 kt	0.5 in up collective	r	3.1	−1.3	0.5	2.4	**2.1**
512	60 kt	1 in down collective	r	−2.3	−3.0	−1.6	0.9	**0.6**
502	60 kt	0.5 in right pedal	r	9.0	12.4	9.1	4.1	**1.0**
503	60 kt	1 in left pedal	r	−11.9	−13.2	−10.5	1.7	**1.5**
401	100 kt	1 in aft cyclic	q	24.1	19.3	17.2	3.9	5.0
706	100 kt	1 in forward cyclic	q	−7.6	−9.8	−10.5	3.1	**2.9**
404	100 kt	0.5 in up collective	r	1.2	−1.4	−0.4	1.5	**0.8**
102	100 kt	1 in right pedal	r	13.5	16.0	14.2	2.4	**2.0**
103	100 kt	1 in left pedal	r	−12.2	−18.5	−14.6	5.5	**2.7**

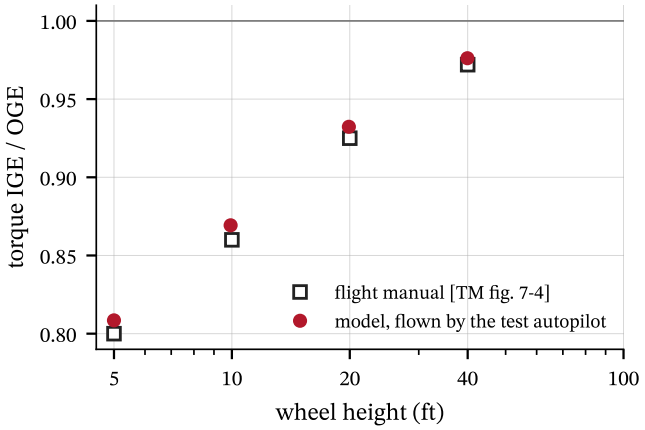


Figure 7: Hover torque in ground effect relative to 100 ft, model (flown by the test autopilot, 16 000 lb, sea level) and the operator's manual hover chart.

simulations also roll the wrong way after a pedal step, which neither reproduces.

5.8 The two tail rotor models

Table 11 collects the comparisons. The blade-element rotor makes more thrust per degree of pitch than Bailey's closed form, some 0.4 in of pedal at low speed, which brings it closer to the test in the hover, and it is about 2 % of pedal travel closer in level flight and at zero sideslip. In crosswinds Bailey's rotor has the smaller pedal error, the blade-element rotor sitting to the right of the high-altitude aircraft, and their tail rotor powers are alike. Bailey's rotor also has the smaller error after pedal steps; both are below GenHel's. The physical differences show at the edges of the envelope: the blade-element rotor reaches its thrust limit from about 158 kt, and in a left crosswind its blades meet the flow reversing through the disc. Flown in FlightGear, a full collective pull from the hover (both engines at 140 % torque, the rotor drooping to 90 %) took it to its pitch stop, stalled at about 1900 lb, and

Table 11: The blade-element and Bailey tail rotors against every test that involves the tail rotor. Pedal in per cent of travel or inches; * not trimmed (the blade-element rotor at its limit).

	test	blade-element	Bailey
hover trim pedal, 5 hovers (in, + right)	−1.05	−1.26	−1.64
level-flight pedal 50–139 kt, mean error (%)	–	−2.6	−4.6
level-flight pedal 162 kt, error (%)	–	−13.2	−9.3
zero-sideslip pedal 60–140 kt, mean error (%)	–	−0.5	−2.4
pedal in crosswinds 10–40 kt, rms (in)	–	0.41	0.36
tail rotor power in crosswinds, mean error (hp)	–	−38	−42
tail rotor power, OGE hover, model/test	–	0.77–0.86	0.77–0.81
yaw rate after 5 pedal steps, mean rms (deg/s; GenHel 3.2)	–	2.0	1.8

the aircraft yawed away against full left pedal; the closed form makes 2400 lb there and holds the heading. The engines' 140 % is itself more than a T700 gives (Section 7). For a pilot the difference is small short of that; for the physics, the blade-element rotor is the one that can tell when it is being asked for too much.

5.9 Stability derivatives

The derivatives (Table 12) explain the weak pitch-up. The model's angle-of-attack stability is close to neutral, as the aircraft's (79-24 found the static stability negative to neutral), but its M_w of 0.0030 is a third of TM-85890's 0.0092: the stabilator's restoring moment nearly cancels the destabilising contributions of rotor and fuselage. Z_w is −0.68 against

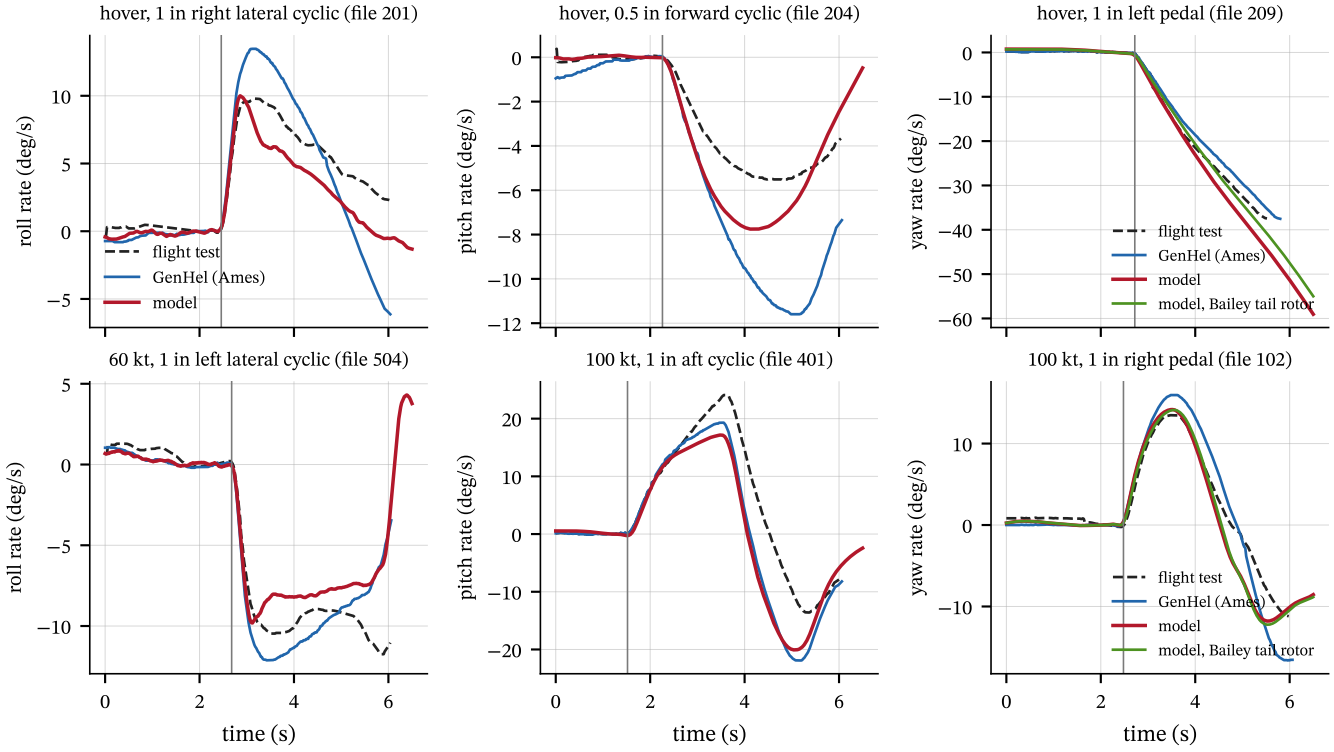


Figure 8: On-axis rate after control steps with the SAS off (TM-88360 app. D): flight test, GenHel as run at NASA Ames, and the model, as increments from trim. The vertical line marks the input.

Table 12: Longitudinal derivatives at 100 kt, 16 400 lb, FS 360.4: TM-85890's model, and this model with the pitch bias actuator and stabilator active, and with them held (rotor and airframe only). Per unit mass or pitch inertia; δ_e inch aft, δ_c inch up.

		TM-85890	model	
	unit	[2]	PBA, stab.	aero only
X_u	1/s	−0.0324	−0.0427	−0.0379
X_w	1/s	0.0643	0.0466	0.0469
X_q	ft/s/rad	2.788	3.576	2.698
Z_u	1/s	−0.0089	0.0406	0.0351
Z_w	1/s	−0.790	−0.677	−0.679
Z_q	ft/s/rad	4.981	−0.101	−1.958
M_u	rad/s/ft	0.0025	0.0022	−0.0026
M_w	rad/s/ft	0.0092	0.0037	0.0030
M_q	1/s	−1.606	−2.825	−1.807
X_{δ_e}	ft/s^2/in	−1.083	−1.265	−1.263
Z_{δ_e}	ft/s^2/in	−6.138	−4.769	−4.815
M_{δ_e}	rad/s^2/in	0.459	0.548	0.541
X_{δ_c}	ft/s^2/in	0.646	0.667	0.680
Z_{δ_c}	ft/s^2/in	−9.630	−8.158	−7.913
M_{δ_c}	rad/s^2/in	0.0951	0.0723	0.1622

−0.79, and a Pitt–Peters wake gives the same, so it is not the inflow model. Z_q has the opposite sign to TM-85890's (−2.0 against 5.0): in hinged-rotor theory pitch rate tilts the disc but does not change its thrust in hub-plane axes, which the model follows. With the SAS on, 79-24 found a long-term aperiodic pitch divergence at 100 kt that the model does not show; it is stable with the pitch bias actuator on and close to neutral with it off.

6 Corrections found during validation

The model is the result of several rounds of validation; the changes that the comparisons forced are listed here because each removed a physical inconsistency rather than tuning a coefficient.

1. *Rotor wake on the airframe in hover.* GenHel puts the in-plane part of the wake's velocity at the fuselage, fin and tail rotor along the body's x axis. The first version used the direction of the in-plane flow instead, which is undefined in hover: a drift of 0.05 ft/s swung 6 ft/s of wash sideways onto the fin and tail rotor and kicked the aircraft in yaw. It now fades to the body axis below 5 ft/s. The hands-off hover with the SAS on became steady (heading within 0.7° in 30 s, against 93° before).
2. *Tip loss in the momentum balance* (Section 5.4): hover power was 4–9 % low and the figure of merit 0.03 high.
3. *Accessory power*: a placeholder of 30 hp was replaced by Sikorsky's 65.8 hp, and an unsourced section drag increment of 0.002 by the 0.0012 of a tripped boundary layer.
4. *Downwash correction terms*: GenHel's empirical wake-swirl forces on the empennage, defined in its motion module rather than the empennage module, were missed at first; without them the aircraft trimmed with 1.4–2.5° of left bank at zero sideslip.
5. *Test conditions*: the control responses were first replayed at the CG of fig. B1. At the CG of table C1, FS 359.5 for the 100 kt cases, the aft-step pitch rate rose from 11.4 to 17.5 deg/s.
6. *Ground effect*: an earlier version scaled thrust; the effect is on the induced velocity, fitted to the hover chart and carried into forward flight with the measured interference ratios.
7. *Ground contact*: wheels alone let the fuselage sink through the ground when the aircraft rolled over, and JSB-

Table 13: Summary: ✓ within the test's scatter, ~ partly, × not reproduced.

Trim controls and attitudes, 50–139 kt	✓
Trim at 162 kt (collective, lateral cyclic, pedal)	~
Heading at full power (blade-element tail rotor)	×
Lateral-directional trim at zero sideslip	✓
Bank change between climb and descent	~
Hover power, OGE and in ground effect	✓
Rotor figure of merit	✓
Tail rotor power in hover	~
Pedal and longitudinal cyclic hovering in a wind, 10–40 kt	~
Tail rotor power hovering in a wind	~
Tail rotor power in level flight: low speed and 150 kt ✓, cruise ×	~
Cruise power (one test ✓, the other ×)	~
Responses to cyclic, collective, pedal in hover and at 60 kt	✓
Yaw damping in hover	~
Pitch-up after an aft step at 100 kt	~
Roll after a pedal step at 100 kt	×
Long-term pitch divergence with SAS (79-24)	×

Sim's wheel force diverges as a strut lies along the ground. Structure contacts from the 3-D model and a rule that takes the wheels out of play beyond 60° of tilt fixed both.

8. *Main rotor wake at the tail*: GenHel gives the tail rotor the empennage's wash, and the first version also turned the in-plane part with the flight path, so that in sideward flight the tail rotor met a crosswind 1.4 times the real one and, through its cant, a phantom climb from 2.1 times the inflow of downwash. The pedal swing between winds from the right and from the left was nearly twice the test's. The ring wake (Section 3) at the tail rotor, fin and stabilator brought the crosswind pedal error from 0.77 to 0.41 in rms and the longitudinal cyclic error from 0.68 to 0.49 in, without changing forward flight.
9. *Trims off the nose*: a trim in angle of attack and sideslip fails at 90° off the nose; the attitudes are now the trim variables there.
10. *Rotor speed and the airframe's yaw*: the rotor speed was integrated relative to the airframe from the torques alone, as if the airframe's yaw did not concern the rotor (Section 5.7). With the term the hover pedal step's yaw-rate error fell from 5.5 to 2.9 deg/s.
11. *Fin blockage*: an intermediate version scaled the tail rotor wake's velocity at the fin to Sikorsky's 20.4 % blockage. That made the fin anti-damping in yaw (+0.06 s^{-1} in the hover, against +0.02 unscaled) and was removed.

7 Assessment

Table 13 summarises. The model reproduces the Black Hawk's trim, its hover and ground-effect power and its short-term responses at least as well as GenHel, from which it was derived, and in hover better. Its open items share one direction: in pitch at speed the model is too stable. A third of TM-85890's M_w, too small a pitch-up after an aft step and no long-term divergence point to the same missing destabilising moment, and Section 5.9 locates it in the balance of rotor, fuselage and stabilator rather than in any single term. The other open items are the cruise power against one of the two tests, the tail rotor's power (low in the hover against one test, high in cruise against another), the hover yaw damping, the anti-torque at the top speed, and the lateral moment per inch of cyclic. At full collective the engines give up to 140 % torque each as the rotor droops, more than a T700 can, and the blade-element tail rotor then stalls. Retreating-blade stall limits the model at high thrust and speed (C_T = 0.009 above μ = 0.28); blade torsion is not modelled, and dynamic stall only as a tail rotor experiment.

8 Conclusions

- A blade-element model of the UH-60A with individual blades, a finite-state dynamic wake, GenHel's airframe and control laws, and engines runs in plain JSBSim, generated from data. It is the flight model of the FlightGear UH-60A.
- Against the US Army flight tests it reproduces level-flight longitudinal cyclic and collective within 9.0 % of travel up to 162 kt, hover power at 0.949–0.979 of the measured over the test's range, ground effect within 0.009 and the SAS-off control responses with a smaller error than GenHel.
- Applying tip loss to the momentum balance as well as the lift brings the rotor's figure of merit and the aircraft's hover power into line with the measurements; GenHel's lift-only tip loss leaves the induced power ideal.
- Where the main rotor's wake goes matters at the tail: GenHel's interference factors, measured with the wake blown aft, put it on the tail rotor and stabilator in any wind. A ring-vortex wake that follows the wind, with the fin in the tail rotor's own wake, halves the error of the pedal hovering in winds from every direction without changing forward flight.
- A blade-element tail rotor reproduces the tests at least as well as GenHel's closed form and shows its thrust limit from about 158 kt, where the closed form has none; that the aircraft flew 162 kt with less tail rotor pitch points to anti-torque the model lacks at the top speed.
- The rotor's angular momentum about its shaft is its own: a rotor speed taken relative to the airframe has to follow the airframe's yaw. The term was missing, and with it the yaw after a hover pedal step comes close to the test's.
- The pitch stability at speed is too strong and the cruise power against the 77-17 data too high; both are documented, not tuned away.

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